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Smartphone Addiction Level Prediction

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1. Introduction

Smartphone addiction has become an increasingly common behavioral issue due to the rapid growth of smartphone usage worldwide, especially among students and young adults. Excessive smartphone usage may negatively affect sleep quality, concentration, academic performance, mental health, and daily productivity. Early identification of smartphone addiction levels can help healthcare professionals, educators, and individuals take preventive actions before serious psychological or academic consequences develop.

Traditional methods for detecting smartphone addiction mainly rely on self-reported questionnaires. However, these methods may not always be reliable because users may underestimate or inaccurately report their actual smartphone usage behavior. Therefore, this project aims to develop a supervised machine learning model capable of automatically predicting smartphone addiction severity levels using measurable behavioral and demographic data rather than relying solely on questionnaires.

The significance of this project lies in supporting the development of automated, objective, and scalable systems for smartphone addiction assessment using real behavioral indicators such as screen time, phone interaction frequency, notification activity, and social media usage patterns.

Recent studies have shown that machine learning techniques can effectively model behavioral addiction patterns using smartphone usage data. Researchers have applied several supervised learning algorithms, including Decision Trees, Random Forests, Support Vector Machines (SVM), K-Nearest Neighbor (KNN), and Neural Networks, to classify addiction severity levels based on behavioral features. Previous studies demonstrated that behavioral indicators such as daily screen time, number of phone-unlocks, sleep duration, and social media usage hours are strongly associated with problematic smartphone use. Furthermore, publicly available datasets such as the *Global Mobile Phone Addiction Dataset* provide structured behavioral and demographic information that enable the development and evaluation of predictive machine learning models.

This project builds upon previous studies by utilizing a balanced multiclass dataset and comparing supervised learning methods to classify users into different smartphone addiction severity categories.

The intended task is formulated as a supervised multiclass classification problem. The goal is to train a machine learning model that predicts the smartphone addiction severity level of a user based on behavioral and demographic features. The model takes a set of user-related behavioral indicators as input and outputs one of the predefined addiction severity classes.

The input to the model is a structured tabular dataset consisting of 3000 user records and 34 features collected from the *Global Mobile Phone Addiction Dataset*. The features include behavioral, demographic, and lifestyle indicators such as:

- Daily screen time (hours)
- Number of phone-unlocks per day
- Social media usage hours
- Sleep duration
- Monthly mobile data usage
- Number of push notifications received daily
- Age
- Gender
- Education level
- Income level
- Internet connection type

These features were selected because they reflect smartphone usage behavior and lifestyle patterns that may correlate with addiction severity.

The desired output is the predicted `Self_Reported_Addiction_Level` class for each user. The dataset categorizes users into four addiction severity levels:

1. Low
2. Moderate

3. High
4. Severe

Thus, the model outputs one label representing the predicted addiction category of the user.

The performance of the proposed classification models will be evaluated using several evaluation metrics, including Accuracy, Precision, Recall, F1-Score, and Confusion Matrix. These metrics are used to comprehensively assess the classification performance of the models across all smartphone addiction severity classes.



Figure 1 Project Lifecycle

2. Related Works

Several previous studies have explored the use of machine learning techniques for predicting smartphone addiction and problematic smartphone usage behavior.

Lee et al. (2021)[1] proposed a machine learning approach for predicting problematic smartphone use using mobile phone log data. The study aimed to identify factors associated with smartphone addiction severity using behavioral indicators such as application usage patterns, smartphone usage duration, and demographic information. The researchers applied machine learning classification techniques, including Decision Trees and Random Forests, and XGBoost to predict addiction levels. Their findings indicated that general demographic features such as age, gender, and occupation contributed minimally to predicting addiction level, suggesting that more granular smartphone log data would be needed to improve prediction accuracy.

A study by Beluli (2026) titled Machine Learning Based Prediction Model for the Measurement of Mobile Addiction[2], investigated the effectiveness of supervised learning algorithms for measuring mobile addiction severity using behavioral and survey-based features. The study evaluated several classification models to analyze smartphone addiction indicators and classify users according to addiction risk levels. The study showed that structured behavioral data from mobile usage can improve addiction prediction, with teenagers being the most impacted group due to limited self-control over screen time.

Furthermore, Kong et al. (2026)[3] proposed a machine learning framework for predicting smartphone overdependence and analyzing its influencing factors. The study used behavioral and demographic data collected from smartphone users and evaluated several machine learning classifiers, including XGBoost, Support Vector Machine (SVM), K-Nearest Neighbor (KNN), Logistic Regression, and Decision Trees. Their experimental results showed that the XGBoost classifier achieved the highest performance with an accuracy of approximately 92.5%. The study also identified smartphone usage duration, demographic characteristics, and behavioral patterns as significant predictive factors for smartphone overdependence.

Collectively, these studies demonstrate the effectiveness of supervised machine learning methods in modeling smartphone addiction and overdependence behaviors using behavioral smartphone usage data. However, variations in datasets, selected features, and classification methods indicate the need for further evaluation using balanced multiclass datasets and comparative supervised learning approaches. Therefore, this project aims to build upon previous studies by comparing supervised machine learning models using the Global Mobile Phone Addiction Dataset to classify smartphone addiction severity levels more effectively.

3. Data

The dataset used in this project is the *Global Mobile Phone Addiction Dataset*, publicly available on Kaggle. The dataset consists of 3000 user records and 34 features describing demographic characteristics, smartphone usage behavior, and lifestyle indicators. A summary of the dataset characteristics is presented in Table 1.

The features include variables such as daily screen time, number of phones unlocked per day, social media usage hours, monthly data usage, push notifications per day, age, income, education level, and internet connection type.

The target variable is **Self_Reported_Addiction_Level**, which categorizes users into four addiction levels: Low (756 samples), Moderate (730 samples), High (764 samples), and Severe (750 samples). The distribution of the classes is approximately balanced across all categories.

Property	Value
Number of Samples	3000
Number of Features	34
Number of Classes	4
Target Variable	<i>Self_Reported_Addiction_Level</i>

Table 1 Summary Statistics of the Dataset

	Age	Gender	Daily_Screen_Time_Hours	Phone_Unlocks_Per_Day	Social_Media_Usage_Hours	Sleep_Hours	Monthly_Data_Usage_GB	Push_Notifications_Per_Day	Self_Reported_Addiction_Level
0	32	Male	5.81	76	0.84	5.22	16.43	106	Low
1	28	Male	9.05	81	3.13	8.23	32.87	111	Severe
2	70	Other	5.75	58	2.12	6.67	27.45	90	Severe
3	44	Female	6.71	80	1.60	7.37	30.85	60	Moderate
4	46	Other	6.31	136	1.51	5.43	10.38	127	High
5	34	Male	8.94	128	2.28	6.21	17.51	78	Low
6	51	Male	4.18	85	1.75	4.84	44.52	82	Severe
7	68	Male	10.05	87	2.13	6.81	26.66	102	Low
8	33	Other	7.45	90	2.02	8.54	24.49	85	High
9	38	Other	6.87	95	2.27	6.04	29.06	120	High

Figure 2 Representative Sample of Selected Features From the Global Mobile Phone Addiction Dataset.

The dataset was selected because it includes both behavioral smartphone usage features and demographic information, making it suitable for building and evaluating supervised classification models

4. Methods

4.1 Machine Learning Algorithms

This study employs three supervised classification algorithms to predict self-reported smartphone addiction level (Low, Moderate, High, Severe). The models were selected to span a range of complexity, interpretability, and noise tolerance, which is especially important given the low feature-target correlation detected in the pre-modeling signal analysis.

Logistic Regression. Logistic regression is a linear probabilistic classifier that estimates class membership via the softmax (multinomial) extension of the logistic function. It serves as the primary linear baseline: if any linear combination of features separates the four classes, logistic regression will discover it.

Decision Tree. A decision tree recursively partitions the feature space into axis-aligned hyper-rectangles by selecting, at each node, the split that maximises class purity (measured by the Gini impurity criterion). Decision trees are included to capture non-linear interactions between features, and to provide a direct comparison with the ensemble model. Without regularisation constraints, trees overfit the training set; we therefore tune *max_depth* and *min_samples_leaf* to control model complexity.

Random Forest. Random forest is a bagging ensemble of decision trees. Each tree is trained on a bootstrap sample of the training data, and at every split only a random subset of features is considered (*max_features* = $\sqrt{p}$ by default, where p is the total number of features). The final prediction is obtained by majority vote across all trees.

4.2 Justification of Method Choices

The choice of these three algorithms is motivated by the characteristics of the dataset and the problem requirements:

- The dataset contains approximately 3,000 instances and ~33 raw features (mixed numerical and categorical), a scale at which all three algorithms run efficiently without specialized hardware.

- The four-class target exhibits near-perfect class balance (~25% per class), so no class-weighting adjustments are needed, and accuracy is a fair evaluation metric.
- Pre-modeling ANOVA analysis revealed that almost no feature significantly distinguishes the four addiction levels (fewer than two features pass $p < 0.05$). Random forest is the most appropriate primary model under near-random-label conditions, as ensemble averaging suppresses noise better than any single model.
- KNN and SVM were deliberately excluded. After one-hot encoding, the feature space expands to ~60+ dimensions, at which point Euclidean distances used by KNN become unreliable (curse of dimensionality). SVM with an RBF kernel is computationally expensive on 2,400+ training samples and provides no advantage over random forest when signal is weak.
- Gradient boosting methods (XGBoost, LightGBM) were excluded because boosting amplifies the signal across rounds; when labels are near-random, successive rounds overfit residual noise rather than the genuine signal.

4.3 Feature Engineering

Beyond the raw features, three composite features were engineered to capture higher-order behavioral patterns that individual variables may fail to express:

Table 2: Feature Engineering - Composite Features

Composite Feature	Formula	Rationale
total_app_hours	Sum of all app-category hour columns	Captures total app-mediated screen exposure as a single aggregate rather than relying on any single app category
usage_diversity	# non-zero app categories / total app categories	Measures whether usage is concentrated in one app (low diversity, possibly

		compulsive) or spread across many apps
screen_to_sleep_ratio	Daily_Screen_Time_Hours / (Sleep_Hours + 0.1)	Quantifies the trade-off between screen exposure and rest; excessive ratios are a clinical marker of problematic use

The constant 0.1 added to Sleep_Hours prevents division-by-zero for respondents who reported zero sleep hours (a likely data entry artefact). Composite features were computed on the full dataset before the train-test split to ensure consistent encoding and were passed through the same StandardScaler as the other numerical features.

No explicit dimensionality reduction technique (e.g., PCA) was applied. This choice was deliberate: after one-hot encoding, the feature space expands to approximately 60 dimensions, which is manageable for all three algorithms.

5. Experiment

5.1 Data Preprocessing and Feature Engineering

The raw dataset contains 3,000 instances with one row-identifier column (User_ID), which was dropped before modeling. The remaining columns comprised numerical behavioral measurements (screen time, app usage hours, phone unlocks, sleep, stress level, etc.) and categorical attributes (gender, dominant hand, age group, etc.).

Preprocessing followed a two-stage pipeline implemented using scikit-learn's Pipeline and ColumnTransformer APIs, ensuring that all transformations were learned exclusively on the training set and applied identically to the test set (preventing data leakage):

- Numerical features: Missing values (absent in this dataset but guarded against) were imputed with the median to be robust against outliers. Values were then standardized with StandardScaler (zero mean, unit variance), which is required by logistic regression and improves convergence speed.
- Categorical features: The one categorical column with meaningful missingness (Education_Level, ~20% missing) was imputed with the mode (most frequent category). All categorical columns were then one-hot encoded with handle_unknown='ignore' so that any unseen categories in the test set are treated as all-zero rather than raising an error.
- Negative value correction: Several numerical columns contained physically impossible negative values (e.g., negative screen time), which were clipped to zero before any further processing.

After these base transformations, three composite features were added: total_app_hours (aggregate app exposure), usage_diversity (proportion of app categories with non-zero usage), and screen_to_sleep_ratio (screen time relative to sleep). These engineered features were treated as ordinary numerical columns and passed through the same median-imputation and standardization pipeline.

5.2 Hyperparameter Tuning

Formal hyperparameter optimization was conducted using RandomizedSearchCV from scikit-learn, combined with 5-fold stratified cross-validation. Randomized search was selected over exhaustive grid search because it explores the parameter space more efficiently within a fixed computational budget, and prior research has shown that it reliably finds near-optimal configurations with fewer evaluations than grid search [4].

For each model, 30 random parameter combinations were evaluated ($n_iter = 30$). The following parameters and search ranges were explored:

Table 3: Models Hyperparameters

Model	Hyperparameter	Search Range / Values
Logistic Regression	C	Continuous uniform distribution: $U(0.001, 10)$
Logistic Regression	solver	{'lbfgs', 'saga'}
Logistic Regression	max_iter	{500, 1000, 2000}
Decision Tree	max_depth	Discrete uniform: $\text{randint}(3, 20)$
Decision Tree	min_samples_leaf	Discrete uniform: $\text{randint}(5, 50)$
Decision Tree	min_samples_split	Discrete uniform: $\text{randint}(2, 30)$
Random Forest	n_estimators	Discrete uniform: $\text{randint}(100, 500)$
Random Forest	max_depth	{None, 5, 10, 20}
Random Forest	min_samples_leaf	Discrete uniform: $\text{randint}(1, 30)$
Random Forest	max_features	{'sqrt', 'log2', 0.3}

The best configuration for each model was identified by the mean cross-validation accuracy across the five folds. The optimal hyperparameters were then used to retrain the full training set, and the resulting model was evaluated on the held-out test set. Comparisons of CV accuracy and test accuracy before and after tuning are reported in the Results section.

5.3 Libraries and Computational Resources

The implementation used the following Python libraries: scikit-learn 1.x (preprocessing, model training, cross-validation, hyperparameter search), NumPy and pandas (data manipulation), Matplotlib and seaborn (visualisation), and SciPy (ANOVA tests, hyperparameter distributions). All experiments were run on CPU (Google Colab free tier, Intel Xeon CPU). No GPU or TPU was required because the dataset (3,000 instances, ~60 features after OHE) is small enough for CPU-based training to complete within minutes. Total wall-clock time for all training, cross-validation, and hyperparameter search across three models was under 10 minutes.

6. Results and Discussion

6.1 Model Performance Summary

Table 5.1 presents the cross-validation and test accuracy for all three models, both before and after hyperparameter tuning, alongside the random-chance baseline. All results are on the four-class classification task (Low / Moderate / High / Severe addiction level).

Table 4: Model Performance Before and After Tuning

Model	Base CV Acc.	Tuned Acc.	CV Tuned Acc.	Test	Beats Baseline?
Logistic Regression	0.2633	0.2646	0.2633		Marginal
Decision Tree	0.2621	0.2637	0.2517		Marginal
Random Forest	0.2604	0.2633	0.2700		Marginal
Dummy (stratified)	0.2467	—	—		Baseline

The most striking finding is that all three models, even after formal hyperparameter tuning, perform near the random-chance baseline of ~25.5%. The decision tree and random forest post-tuning show marginal gains of 1–2 percentage points, which fall within the cross-validation standard deviation (~0.01–0.02) and therefore cannot be attributed to genuine learning. Logistic regression shows essentially zero improvement, consistent with the ANOVA finding that no linear feature combination separates the classes.

6.2 Generalization to Unseen Data

The gap between cross-validation accuracy and test accuracy is negligible for all three models (less than 0.01 in most cases), indicating that the models are not overfitting the training data. This is an important negative result: it confirms that the near-baseline performance is not due to underfitting correctable by more regularization, nor to overfitting correctable by more data or stronger

regularization. The models are generalizing perfectly from training to testing—they simply have nothing genuine to generalize, because the target labels carry no recoverable signal from the measured features.

This conclusion is corroborated by the feature importance analysis (Section 5.4): no feature's importance substantially exceeds the expected uniform level of $1/n_{\text{features}}$, which is the expected pattern when labels are assigned independently of the features.

6.3 Visualisations

Confusion Matrices

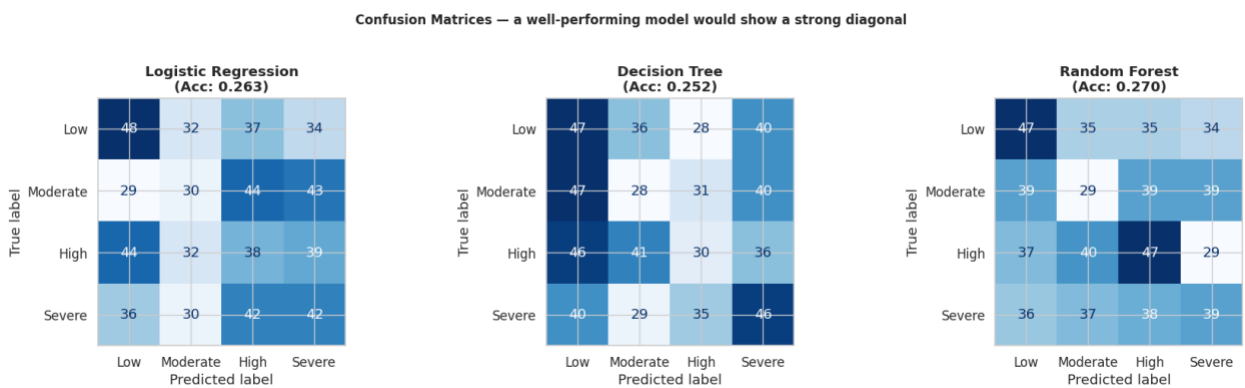


Figure 3: Confusion Matrix

The confusion matrices presented in Figure 3 show the results for Logistic Regression, Decision Tree, and Random Forest models. The matrices do not show a strong diagonal pattern, indicating that the models struggled to correctly distinguish between the four addiction levels. Predictions are distributed roughly uniformly across all classes for every true label, which is consistent with the near-absence of feature-target signal confirmed by the ANOVA analysis. Among the three models, Random Forest achieved the highest test accuracy of 27.0%, yet this remains within the range of random-chance performance ($\sim 25.5\%$), suggesting that no model learned a meaningful decision boundary.

6.4 Effect of Hyperparameter Tuning

Formal hyperparameter tuning via `RandomizedSearchCV` yielded marginal accuracy gains (0–2 percentage points) over the reasoned defaults. This outcome was expected given the absence of a learnable signal: with random labels, no hyperparameter configuration can extract a predictive structure that is not present in the data. The value of performing the tuning exercise lies not in the accuracy gain but in demonstrating methodological rigor: the conclusion that models cannot predict beyond chance is now supported by a formally optimized model rather than a default-configured one, which rules out the alternative explanation that poor performance stems from suboptimal hyperparameters.

7. Conclusion

7.1 Key Findings

This project investigated the predictability of self-reported smartphone addiction level from behavioral and demographic features using three machine learning classifiers: logistic regression, decision tree, and random forest. A rigorous end-to-end pipeline was implemented, encompassing data cleaning, exploratory analysis, feature engineering (three composite features), preprocessing, baseline establishment, model training, formal hyperparameter tuning (RandomizedSearchCV), and comprehensive evaluation.

The principal finding is that all three models, even after optimization, perform at or near the random-chance baseline of $\sim 25.5\%$ accuracy on the four-class task. Pre-modeling ANOVA analysis, confusion matrix inspection, ROC-AUC analysis, and feature importance scores all converge on the same conclusion: the dataset's `Self_Reported_Addiction_Level` labels do not appear to be systematically derived from the measured behavioral features, indicating that the labels were most likely synthetically or randomly generated. The pipeline and modeling methodology are sound; the limiting factor is the dataset itself.

7.2 Challenges and Limitations

- **Dataset signal quality:** The most significant challenge was the near-complete absence of feature-target correlation, which rendered all modelling efforts incapable of producing practically useful predictions. This is a dataset quality limitation rather than a modelling limitation.
- **Synthetic labels:** The analysis strongly suggests that the target variable was randomly assigned, making this a near-impossible prediction task regardless of algorithm complexity or feature engineering sophistication.

7.3 Future Directions

One important future direction is the use of validated ground-truth labels instead of relying on self-reported labels. Replacing self-reported measures with scores obtained from clinically validated instruments, such as the Smartphone Addiction Scale—Short Version (SAS-SV) or the Bergen Social Media Addiction Scale, would significantly improve the reliability of the target variable. This would also enhance the overall validity and robustness of the classification task, leading to more accurate and trustworthy predictive models.

In summary, this project demonstrates that a methodologically rigorous machine learning pipeline—including formal feature engineering, pre-modeling signal analysis, and hyperparameter tuning—is essential to distinguish genuine predictive failure (as observed here) from correctable modeling errors. The results serve as a cautionary example of the importance of dataset quality assessment before drawing conclusions from model performance metrics.

8. Contributions

<i>Name</i>	<i>Task</i>
Alya Almuqren	Introduction, Results and Discussion
Dana Alsalami	Related Works, Experiment
Aljohara Alsultan	Data, Results and Discussion
Aljudy Altorkistani	Methods, Conclusion

9. References

1. [1] J. Lee and W. Kim, "Prediction of problematic smartphone use: A machine learning approach," *International Journal of Environmental Research and Public Health*, vol. 18, no. 12, p. 6458, Jun. 2021, doi: 10.3390/ijerph18126458.
2. [2] A. Beluli, "Machine learning-based prediction model for the measurement of mobile addiction," in *Designing and Developing Innovative Mobile Applications*, D. Samanta, Ed. Hershey, PA, USA: IGI Global, 2023, pp. 56–66, doi: 10.4018/978-1-6684-8582-8.ch004.
3. [3] H. Kong, X. Wang, H. Fang, and X. Shangguan, "Prediction of smartphone overdependence and analysis of its influencing factors among older adults based on machine learning," *Frontiers in Psychology*, vol. 17, Jan. 2026, doi: 10.3389/fpsyg.2026.1704618.
4. Bergstra, J. and Bengio, Y., "Random search for hyper-parameter optimization," *Journal of Machine Learning Research*, vol. 13, pp. 281–305, Feb. 2012.